A hypermap whose distinguished involutions fail the intersection property

Human Compression project

The unnamed paragraph opening Section 5.4 of Leemans's fixed survey edition [1] states that the hypermaps of Conder–Potočnik– Širáň [2] satisfy the intersection property, citing the absence of nontrivial cyclic normal subgroups and Conder–Oliveros, Corollary 4.2. The latter is the string result reproduced as survey Theorem 3.2: a pair of the three involutions must commute. That condition cannot be discarded. We give an explicit counterexample to the unnamed inference, without questioning the hypermap classification in [2].

Proposition 1. *There is a regular hypermap with full group* $\mathrm{PSL}_2(11)$ *and type* $(6,6,5)$ *whose three distinguished involutions do not form a C-group marking.*

Proof. All matrices below have entries in $\mathbb{F}_{11}$. Put

$$A = \begin{pmatrix} 5 & 1 \\ 7 & 6 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 10 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 4 & 3 \\ 9 & 7 \end{pmatrix}.$$

They have determinant one and trace zero, so their images a, b, c in $\mathrm{PSL}_2(11)$ are involutions. Direct multiplication gives

$$ABACBABCAC = -U, \qquad U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The matrices U and B generate $\mathrm{SL}_2(11)$. Indeed the powers of U give every elementary upper unipotent matrix, and conjugating by B gives every elementary lower unipotent matrix. These perform row elimination. To include the resulting diagonal matrices explicitly, for $s \neq 0$ one has

$$\begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -s^{-1} & 1 \end{pmatrix} \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} s & 0 \\ 0 & s^{-1} \end{pmatrix} B.$$

Thus $\langle a, b, c \rangle = \mathrm{PSL}_2(11)$.

Two more direct products are

$$X = (AB)^3 = \begin{pmatrix} 3 & 10 \\ 10 & 8 \end{pmatrix}, \qquad Z = (BC)^3 = \begin{pmatrix} 1 & 3 \\ 3 & 10 \end{pmatrix}, \qquad XZ = -B.$$

Write x, z for their projective images. Since $x = (ab)^3$ and $x = bz^{-1}$, the element x belongs to $\langle a, b \rangle \cap \langle b, c \rangle$. The displayed matrix X is neither $\pm I$ nor $\pm B$. Therefore

$$\langle a, b \rangle \cap \langle b, c \rangle \supsetneq \langle b \rangle,$$

which is the required failure of the intersection property. The group $\mathrm{PSL}_2(11)$ is nonabelian simple, so it has no nontrivial cyclic normal subgroup. Here we retain the standard simplicity theorem for $\mathrm{PSL}_2(q)$ with $q \geq 4$; no classification search is needed to infer simplicity from a group order.

Finally put $r = ab$ and $s = bc$. Their orders are six and six, while $rs = ac$ has order five; these orders follow by multiplying the displayed matrices and checking powers modulo $\{I, -I\}$.

The equation $xz = b$ gives $b = r^3s^3$, so r, s generate the same group as a, b, c. Conjugation by the involution b inverts both r and s. The type is hyperbolic, since $1/6 + 1/6 + 1/5 < 1$. The introduction of [2] identifies exactly this situation as an inner-reflexible generating triple: it gives a regular hypermap on a nonorientable surface with full automorphism group $\langle r, s\rangle$. This establishes the hypermap interpretation as well as the group calculation. □

The valid string theorem remains unchanged. Hypermap enumeration can still provide candidates for C-group markings, but the appropriate intersection tests must be imposed. The blanket transfer made in the survey is false. The additional literal assertion about all values of q also needs its field range: $\mathrm{PSL}_2(2) = S_3$ has a normal cyclic subgroup of order three. The main hyperbolic scope of [2] already excludes that elliptic example. Our counterexample has $q = 11$ and lies within that hyperbolic scope, so it is independent of this small-field qualification.

The displayed certificate was found while auditing this particular survey clause. This note does not claim to settle a named open conjecture, to replace the full hypermap enumeration, or to complete the survey's other numerical obligations. The hypermap correspondence and original enumeration remain attributed to [2].

References

[1] D. Leemans, *String C-group representations of almost simple groups: a survey*, arXiv:1910.08843v2, 11 May 2020, unnamed paragraph at the start of Section 5.4.

[2] M. Conder, P. Potočnik and J. Širáň, *Regular hypermaps over projective linear groups*, J. Aust. Math. Soc. 85 (2008),155–175, especially pp. 156–157 and Section 6; doi:10.1017/S1446788708000827.